

The ten helical twist angles of B-DNA

W.Kabsch, C.Sander and E.N.Trifonov

**Junguk and Ram
School of Informatics
Indiana University, Bloomington**

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Purpose

- Calculate 10 helical twist angles of B-DNA based on the previous experimental data

Introduction

- Twist angles for different combinations of base-pairs in the DNA molecules are **not identical**
- DNA molecule – linear array of 10 possible stereo-chemically different combinations of adjacent base-pairs
- No end-effect involved

Introduction

- The twist angles between A.T base-pairs in poly(dA).poly(dT) and G.C base-pairs in poly(dG).poly(dC) were experimentally measured

Introduction

- Experimental Methods
 - Electrophoretic band-shift method
 - Crystallography
 - Nuclease digestion of DNA adsorbed to mica or Ca-phosphate surface

Introduction

- Twist angles for 33 short DNA fragments with different nucleotide sequences measured
- Six out of ten are measured - AA, AT, CC, CG, GA, GC
- The values obtained indicated that twist angles for different combinations of base-pairs in the DNA molecule are not identical.
- Structure of the self-complementary dodecanucleotide 5' -CGCGAATTCGCG-3'

Introduction

- Assumption - the twist angles between adjacent base-pairs in the DNA molecule are additive.
- A linear system of 40 equations was derived.
- From these equations, twist angles of AC, AG, CA, TA are estimated.
- Equations - Statistically consistent, proved that the angle between any particular two base-pairs is a function of only these base-pairs, independent of nearest neighbours.

Method

- Linear solution
 - **10 unknowns (dinucleotides)**
 - **Least square : Minimizing the sum of the square of the difference**
- system of 40 equations
 - In fact, 36 statistically consistent equations were used.
 - Minimal 10 non-singular equations are needed

Method

- How to derive equations

12 bases deleted
 TAAGCT|TTAATGCGGTAG|TTTATCA.... pBR322
 TAAGCT|TATA.....TATAT|TTTATCA pAT29
 41 bases inserted

Lost twist angles: $3\tau_{AA(TT)}$, $2\tau_{AC(GT)}$, τ_{AG} , τ_{AT} , $\tau_{CA(TG)}$, $\tau_{CC(GG)}$,
 τ_{CG} , τ_{GC} , $2\tau_{TA}$

New angles : $2\tau_{AA(TT)}$, $20\tau_{AT}$, $20\tau_{TA}$

Difference : $-\tau_{AA} - 2\tau_{AC} - \tau_{AG} + 19\tau_{AT} - \tau_{CA} - \tau_{CC} - \tau_{CG} - \tau_{GC} + 18\tau_{TA}$

Table I. System of equations for determination of individual twist angles

Eq.No	AA	AC	AG	AT	CA	CC	CG	GA	GC	TA	τ_e	ϕ_e	τ_c	$\tau_e - \tau_c$	$\frac{\tau_e - \tau_c}{\sigma_e}$	Components of σ_e			Plasmids compared	Ref
																σ_1	σ_2	σ_3		
1	7	8	5	6	7	9	2	5	2	6	33.96	0.15	33.93	0.03	0.2	0.06	0.10	0.09	pTR161/pTR190	1
2	6	6	5	6	6	8	2	5	2	6	33.64	0.15	33.89	-0.25	-1.7	0.06	0.09	0.10	pTR161/pTR182	1
3	4	3	2	5	3	6	2	4	1	4	33.64	0.18	33.96	-0.32	-1.8	0.06	0.09	0.15	pTR161/pTR188	1
4	3	2	2	3	1	5	2	3	1	3	33.64	0.24	33.85	-0.21	-0.9	0.06	0.09	0.21	pTR161/pTR183	1
5	3	2	2	3	1	5	1	3	1	3	33.96	0.25	34.02	-0.06	-0.2	0.06	0.10	0.22	pTR161/pTR199	1
6	3	4	3	3	5	2	0	2	1	3	33.96	0.20	33.93	0.03	0.2	0.06	0.10	0.16	pTR183/pTR182	1
7	0	0	0	0	0	0	1	0	0	0	35.29	4.20	29.93	5.36	1.3	0.07	0.10	4.20	pTR199/pTR183	1
8	1	2	0	0	1	1	0	0	0	0	36.00	0.85	34.33	1.67	2.0	0.07	0.11	0.84	pTR182/pTR190	1
9*	1	1	0	2	2	1	1	1	0	1	36.00	0.44	33.84	2.16	4.9	0.07	0.11	0.42	pTR199/pTR188	1
10*	16	0	0	1	1	0	0	1	0	0	36.36	0.13	35.43	0.93	7.1	0.07	0.11		pLP119/pBR322	2
11	34	0	0	2	2	0	0	0	1	1	35.64	0.13	35.49	0.15	1.2	0.07	0.11		pLP140/pBR322	2
12	41	0	0	1	1	0	0	1	0	0	35.64	0.13	35.54	0.10	0.8	0.07	0.11		pLP144/pBR322	2
13	0	1	1	0	2	14	0	0	1	0	33.96	0.12	33.79	0.17	1.5	0.06	0.10		pLP219/pBR322	2
14	0	1	1	0	2	17	0	0	1	0	33.64	0.11	33.77	-0.13	-1.2	0.06	0.09		pLP222/pBR322	2
15	0	0	0	0	0	23	1	0	1	0	33.96	0.12	33.76	0.20	1.7	0.06	0.10		pLP225/pBR322	2
16	0	0	0	0	0	32	1	0	1	0	33.64	0.11	33.73	-0.09	-0.8	0.06	0.09		pLP234/pBR322	2
17	7	0	0	-1	-1	0	0	1	-1	-1	36.00	0.13	36.04	-0.04	-0.3	0.07	0.11		pLP140/pLP142	2
18	18	0	0	1	1	0	0	-1	1	1	35.29	0.12	35.54	-0.25	-2.0	0.07	0.10		pLP119/pLP140	2
19	25	0	0	0	0	0	0	0	0	0	35.64	0.13	35.62	0.02	0.2	0.07	0.11		pLP119/pLP144	2
20*	0	-1	-1	0	-2	9	1	0	0	0	32.73	0.11	33.68	-0.95	-8.8	0.06	0.09		pLP219/pLP225	2
21	0	-1	-1	0	-2	15	1	0	0	0	33.64	0.11	33.67	-0.03	-0.2	0.06	0.09		pLP222/pLP234	2
22	0	-1	-1	0	-2	18	1	0	0	0	33.64	0.11	33.66	-0.02	-0.2	0.06	0.09		pLP219/pLP234	2
23*	0	0	0	0	0	12	0	0	0	0	32.73	0.08	33.65	-0.92	-8.5	0.06	0.09		pLP219/pLP222	2
																			& pLP225/pLP234	
24	43	37	40	28	62	46	24	47	33	12	34.29	0.09	34.10	0.19	2.2	0.07	0.05	0.01	p372/pBR322	3
25	11	8	7	3	12	5	1	7	3	2	34.12	0.12	34.16	-0.04	-0.3	0.06	0.05	0.09	p60/pBR322	3
26	9	15	11	2	16	12	12	12	12	2	33.96	0.09	34.08	-0.12	-1.3	0.06	0.05	0.05	p104/pBR322	3
27	20	0	0	1	2	0	0	0	1	0	35.47	0.09	35.56	-0.09	-1.0	0.07	0.05		pAA24/pBR322	3
28	85	0	-1	0	-1	0	0	0	-1	0	35.64	0.09	35.67	-0.03	-0.4	0.07	0.05		pAA82/pBR322	3
29	0	0	2	20	0	0	0	0	1	21	33.64	0.08	33.66	-0.02	-0.2	0.06	0.05		pAT44/pBR322	3
30	-1	-2	-1	19	-1	-1	-1	0	-1	18	33.64	0.08	33.69	-0.05	-0.7	0.06	0.05		pAT29/pBR322	3
31	1	0	2	34	0	0	0	0	1	35	33.80	0.08	33.72	0.08	1.1	0.06	0.05		pAT73/pBR322	3
32	65	0	-1	-1	-3	0	0	0	-2	0	35.82	0.09	35.72	0.10	1.2	0.07	0.05		pAA82/pAA24	3
33	0	0	0	0	0	0	0	0	1	0	42.15	1.70	40.18	1.97	1.2	1.70				4
34	0	0	0	0	0	0	1	0	0	0	32.90	1.70	29.93	2.97	1.7	1.70				4
35	0	0	0	0	0	0	0	1	0	0	39.40	1.70	36.83	2.57	1.5	1.70				4
36	1	0	0	0	0	0	0	0	0	0	36.75	1.70	35.62	1.13	0.7	1.70				4
37	0	0	0	1	0	0	0	0	0	0	32.20	1.70	32.14	0.06	0.0	1.70				4
38	1	0	0	0	0	0	0	0	0	0	36.00	0.36	35.62	0.38	1.1	0.36				5
39	0	0	0	1	0	0	0	0	0	1	34.29	0.34	33.74	0.55	1.6	0.34				5
40	0	0	0	0	0	0	1	0	1	0	34.29	1.00	35.06	-0.77	-0.8	1.00				6

Results

- Absolute difference between experimental and calculated angles
 - **Significantly smaller than measurement error (0.36°)**
- $(\tau_e - \tau_c) / \sigma_e$ have mean zero and unit variance $\Rightarrow$ consistent equations
 - Four equations were exception. Erroneous.
- Synthetic DNA is different from natural one?
 - Practically identical

Table II. Twist angles of B-DNA ($^{\circ}$)

	Angles known before	Ref.	Angles calculated	Final solution (corrected)
τ_{AA}	35.64 ± 0.36	2	35.62 ± 0.07	35.62 ± 0.06
	35.64 ± 0.36	3		
	36.75 ± 1.7	4		
	36.00 ± 0.36	5		
τ_{AC}	—		34.0 ± 1.4	34.4 ± 1.3
τ_{AG}	—		27.9 ± 1.6	27.7 ± 1.5
τ_{AT}	32.2 ± 1.7	4	32.1 ± 1.1	31.5 ± 1.1
τ_{CA}	—		34.4 ± 1.0	34.5 ± 0.9
τ_{CC}	33.64 ± 0.34	2	33.65 ± 0.07	33.67 ± 0.07
τ_{CG}	32.9 ± 1.7	4	29.9 ± 1.1	29.8 ± 1.1
τ_{GA}	39.4 ± 1.7	4	36.8 ± 0.9	36.9 ± 0.9
τ_{GC}	42.2 ± 1.7	4	40.2 ± 1.2	40.0 ± 1.2
τ_{TA}	—		35.3 ± 1.1	36.0 ± 1.1

Discussion

- The system of equations is consistent
 - Twist angle between any two adjacent base-pair are too a good approximation independent of the particular sequence they are included in
- Four twist angles (AC, AG, CA, and TA) are newly estimated

Discussion

- Sequence-dependent variation - probable important structural feature involved in the specific recognition of DNA.
- **Starting** (C or A) tend to be larger than average, while **ending** (C or A) tend to be smaller. -> Cross compensation -> Preventing over, undertwisting.